

FIT9132 Introduction to Databases

Assignment 1 - ReadMore Community Library (RCL)

Assignment 1 Logical Checklist

Required Actions:

- ☐ Carefully checked the Marking Rubric in the assignment document so you are aware of the mark allocation
- ☐ Normalisation
 - ☐ Normalised two forms (Library Catalogue and Borrowers Quarterly Report [two samples are provided for the Library Catalogue but only normalise each type of form once]). Normalised each from UNF to 3NF, showing all stages and dependencies at each step (i.e. partial dependencies in 1NF, transitive dependencies in 2NF, full dependencies in 3NF)
 - ☐ included all candidate keys for each relation at 1NF
 - ☐ Included all attributes shown on the forms
 - ☐ Did *not* add any surrogate keys in normalisation
 - ☐ Indicated PK's using underline of the PK attributes
 - ☐ Completed attribute synthesis
- ☐ Logical Model
 - ☐ Followed the unit logical model notation/requirements:
 - ☐ Used Crow's foot/Information Engineering notation for the logical model
 - ☐ Does *not* show data types and sizes
 - ☐ Does show legend on the logical model
 - ☐ Does show a label for each relationship
 - ☐ Added common prefix for all attributes in each relation (e.g. emp_no, emp_name)
 - ☐ Added comments for all attributes (used comments in RDBMS)
 - ☐ Added at least one surrogate key to the logical model and added an explanation of why chosen to the assumptions document
 - ☐ No relation with more than two attributes in the key remains (if more than two, a surrogate must be added)
 - ☐ Included all required check clauses and lookup tables
 - ☐ Included FULL normalisation results (final 3NF) in the logical model - all relations and attributes
 - ☐ Checked that the relation/attribute names in the normalisation document are identical to the logical model relation/attribute names on your model
 - ☐ For each relation, all attributes are shown on the diagram (no downward pointing diamond displayed to show some attributes are hidden)
 - ☐ Made sure there are no data anomalies in the final logical model (ie. all relations are in 3NF)
- ☐ Schema File
 - ☐ Schema file has extension .sql
 - ☐ Included DROP table commands at the head of the schema file
 - ☐ Has not been edited other than to add a header (student details) and SPOOL/ECHO commands
 - ☐ Captured run of schema file via SPOOL/ECHO commands

Required files have been pushed to the FIT GitLab server (at least 9 pushes of the model):

- ☐ rcl_normalisation source file (.docx in MS Teams) and final pdf on GitLab
- ☐ rcl_logical.pdf (check this is your final model and produced via File - Print Diagram - To PDF File from within Data Modeller, do not use screen capture)
- ☐ rcl_model project folder, which includes:
 - ☐ the .dmd file,
 - ☐ and the folder of the same name

You must push this project folder at least nine times while you are drawing the model with the Oracle Data Modeler

- ☐ rcl_schema.sql
- ☐ rcl_schema_output.txt
- ☐ rcl_assumptions source file (.docx in MS Teams) and final pdf on GitLab
- ☐ rcl_####_diary source file (.docx in MS Teams) and final pdf on GitLab (#### replaced by group number)

Finally, 7 individual files have been submitted to Moodle:

- ☐ rcl_normalisation.pdf
- ☐ rcl_logical.pdf
- ☐ rcl_model.zip, which includes:
 - ☐ the .dmd file,
 - ☐ and the model folder.

You **MUST ensure that the zip archive of the model is tested for completeness, i.e. includes both logical and relational models** (unzipped and opened in a new location - see video: Preparing Files for Submission in ed)

- ☐ rcl_schema.sql
- ☐ rcl_schema_output.txt
- ☐ rcl_assumptions.pdf
- ☐ rcl_####_diary.pdf (#### replaced by group number)